Conditional Probability

* $P(A|B)$ — not — $= P(A-B) = P(A \cap B^c) \rightarrow$ in some refs.

$P(A/B)$: Prob. of A given that B.

* $P(A/B) = \dfrac{P(A \cap B)}{P(B)}$, $S = \{1, 2, 3, 4, 5, 6\}$

$A = \{6\}$, $B = \{2, 4, 6\}$

$A \cap B = \{6\}$

$P(A|B) = \dfrac{1/\cancel{6}}{3/\cancel{6}} = \dfrac{1}{3}$

$P(D) = 0.83$, $P(A) = 0.82$, $P(D \cap A) = 0.78$

i) $P(A/D) = \dfrac{P(A \cap D)}{P(D)} = \dfrac{0.78}{0.83} = \_\_\_$

ii) $P(D/A) = \dfrac{P(A \cap D)}{P(A)} = \dfrac{0.78}{0.82} = \_\_\_$

iii) $P(A/D^c) = \dfrac{P(A \cap D^c)}{P(D^c)} = \dfrac{P(A) - P(A \cap D)}{1 - P(D)} = \_\_\_$

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* If A & B are Indep.:

$P(A|B) = P(A)$, $P(B|A) = P(B)$

$P(A|B) = P(A) = \dfrac{P(A \cap B)}{P(B)} \longrightarrow P(A \cap B) = P(A) \cdot P(B)$.

$P(A) \cdot P(B) = 0$

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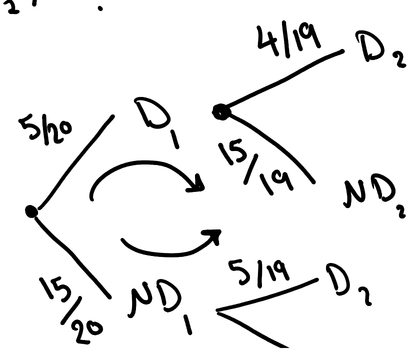
* $P(D_1 \cap D_2) = ?$

2 fuses

5 D
15 ND

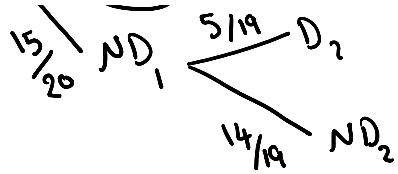
20 fuses

method 1:



$P(D_1 \cap D_2)$

$= \dfrac{5}{20} \cdot \dfrac{4}{19} = \_\_\_$



$= \frac{5}{20} \cdot \frac{4}{19} = \checkmark$

$P(1D \,\&\, 1ND) = \frac{5}{20} \cdot \frac{15}{19} + \frac{15}{20} \cdot \frac{5}{19} = \checkmark$

method 2:

$P(A \cap B) = P(A) \cdot P(B/A)$

$= P(B) \cdot P(A/B)$

$P(D_1 \cap D_2) = P(D_1) \cdot P(D_2/D_1) = \frac{5}{20} \cdot \frac{4}{19} = \checkmark$

$P(1D \,\&\, 1ND) = P(D_1 \cap ND_2) + P(ND_1 \cap D_2) = P(D_1) \cdot P(ND_2/D_1) + P(ND_1) \cdot P(D_2/ND_1)$

$= 5/20 \cdot \frac{15}{19} + \frac{15}{20} \cdot \frac{5}{19} = \checkmark$

method 3:

$P(D_1 \cap D_2) = \frac{{}^5C_2}{{}^{20}C_2} = \checkmark$

$P(1D \,\&\, 1ND) = \frac{{}^5C_1 \cdot {}^{15}C_1}{{}^{20}C_2} = \checkmark$

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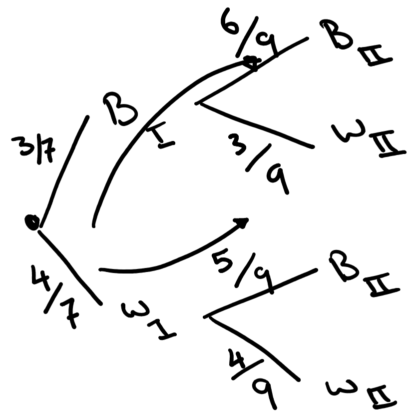

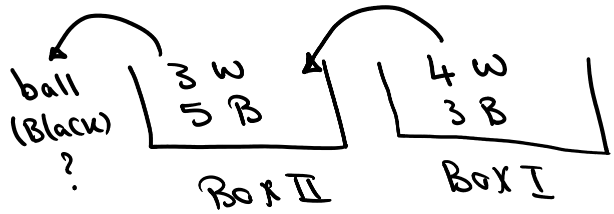

$P(B_{II}) = \frac{3}{7} \cdot \frac{6}{9} + \frac{4}{7} \cdot \frac{5}{9} = \checkmark$

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* $P(A_1 \cap A_2) = P(A_1) \cdot P(A_2/A_1) \longrightarrow$ In general

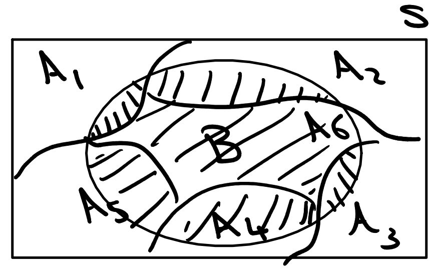
$= P(A_1) \cdot P(A_2) \longrightarrow$ Indep.

* $P(A_1 \cap A_2 \cap A_3) = P(A_1) \cdot P(A_2/A_1) \cdot P(A_3/A_1 \cap A_2) \longrightarrow$ in general.

$= P(A_1) \cdot P(A_2) \cdot P(A_3) \longrightarrow$ indep.

* $P(A_1 \cap A_2 \cap A_3 \ldots \cap A_k) = P(A_1) \cdot P(A_2/A_1) \cdot P(A_3/A_1 \cap A_2) \cdots P(A_k/A_1 \cap A_2 \ldots \cap A_{k-1})$

$= P(A_1) \cdot P(A_2) \cdots P(A_k) \longrightarrow$ indep.



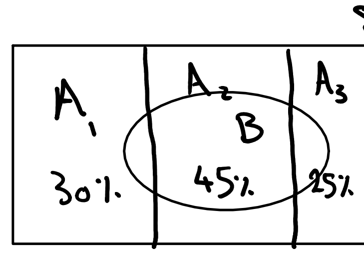
*Partition:

① $A_1, A_2, \dots, A_6$ are mutually Exclusive.

② $A_1 \cup A_2 \cdots \cup A_6 = S$

$\therefore$ $A_1, A_2, \dots,$ and A_n Consist a Partition on S.

$$P(B) = \sum_{i=1}^{6} P(A_i \cap B) = \sum_{i=1}^{6} P(A_i) \cdot P(B/A_i) \longrightarrow$$



$P(D/A_1) = 0.02$

$P(D/A_2) = 0.03$

$P(D/A_3) = 0.02$

$$P(D) = P(D \cap A_1) + P(D \cap A_2) + P(D \cap A_3)$$

$$= P(A_1) \cdot P(D/A_1) + P(A_2) \cdot P(D/A_2) + P(A_3) \cdot P(D/A_3).$$

$$= 0.3 \cdot (0.02) + 0.45 (0.03) + 0.25 (0.02) = \underline{\quad}$$

$$P(A_2/D) = \frac{P(D/A_2) \cdot P(A_2)}{P(D)} = \frac{0.03 \cdot (0.45)}{\underline{\quad}} = \underline{\quad}$$